

Time Horizon of the Effect Size in Meta-Analysis

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The views expresses are those of the authors and do not necessarily reflect the views of the Czech National Bank.

Illustration on: Structural Reform and Growth in Transition

- The effects of reform on growth in developing countries (as a whole) have been at the top of the agenda recently
- Transition economies close to natural experiment: need to reform everything and from scratch
- What does the evidence from the econometric literature on growth in TEs say?

Growth regressions

- General form

$$g = \alpha + \beta R + \delta Z + \epsilon, \quad (1)$$

- Specific forms

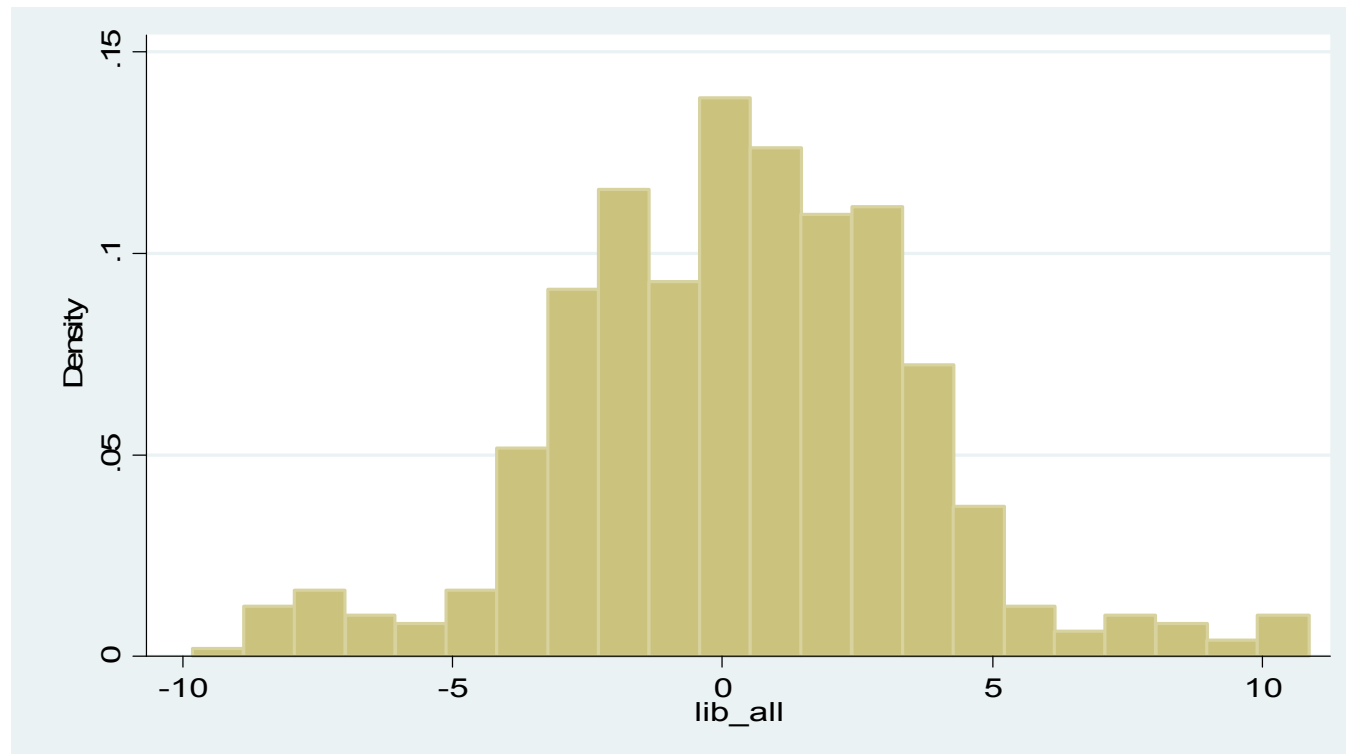
$$g_{it} = \alpha + \beta(R_{it} - R_{it-1}) + \delta R_{it-1} + \gamma Z_{it} + \epsilon_{it}, \quad (2)$$

$$g_{it} = \alpha + \beta R_{it} + \delta R_{it-1} + \gamma Z_{it} + \epsilon_{it}, \quad (3)$$

$$g_{it} = \alpha + \beta R_{it} + \gamma Z_{it} + \epsilon_{it}, \quad (4)$$

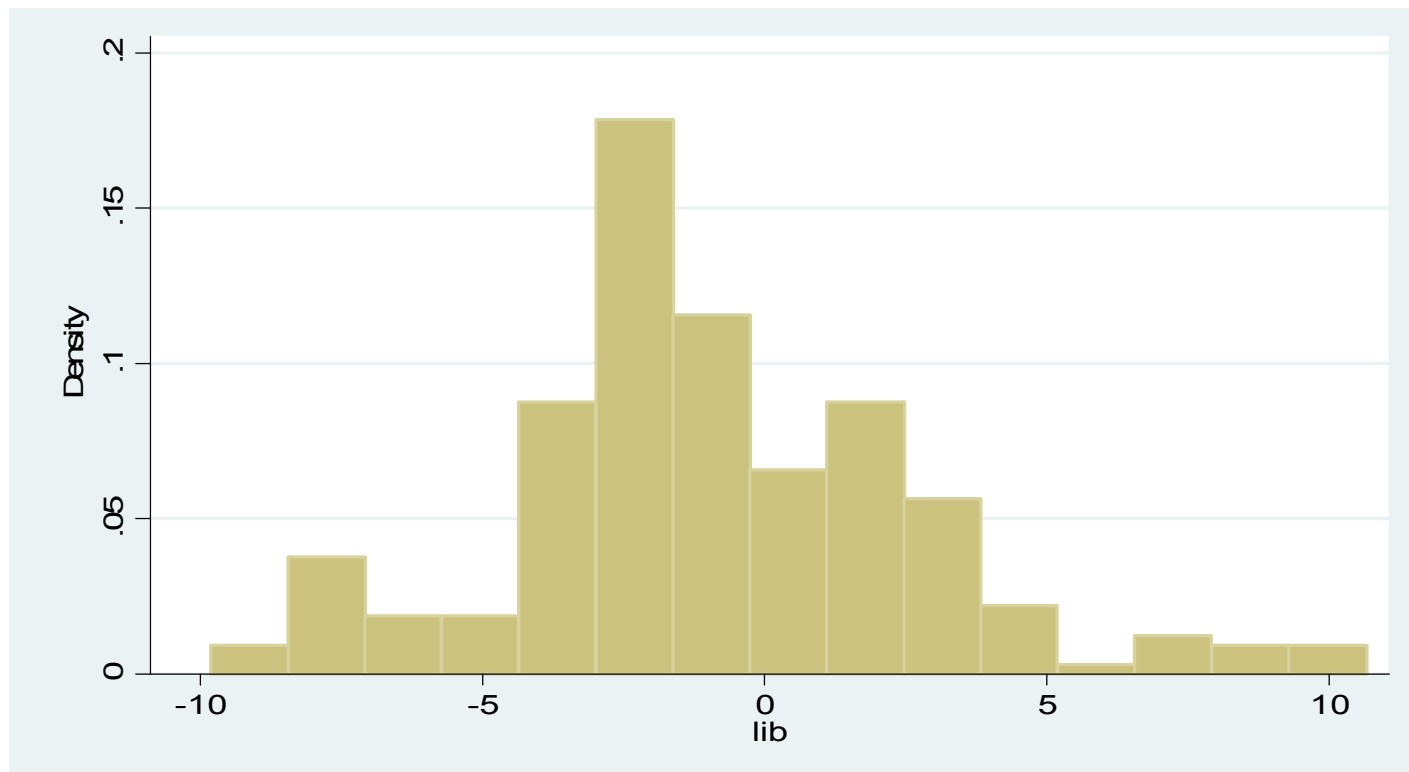
- LR effect: β in (1) ($T > 1y$); δ in (2); $\beta + \delta$ in (3); β in (4)
- SR effect: β in (1) ($T \leq 1y$); β in (3); β in (2)

Link between structural reform and growth: overall effect



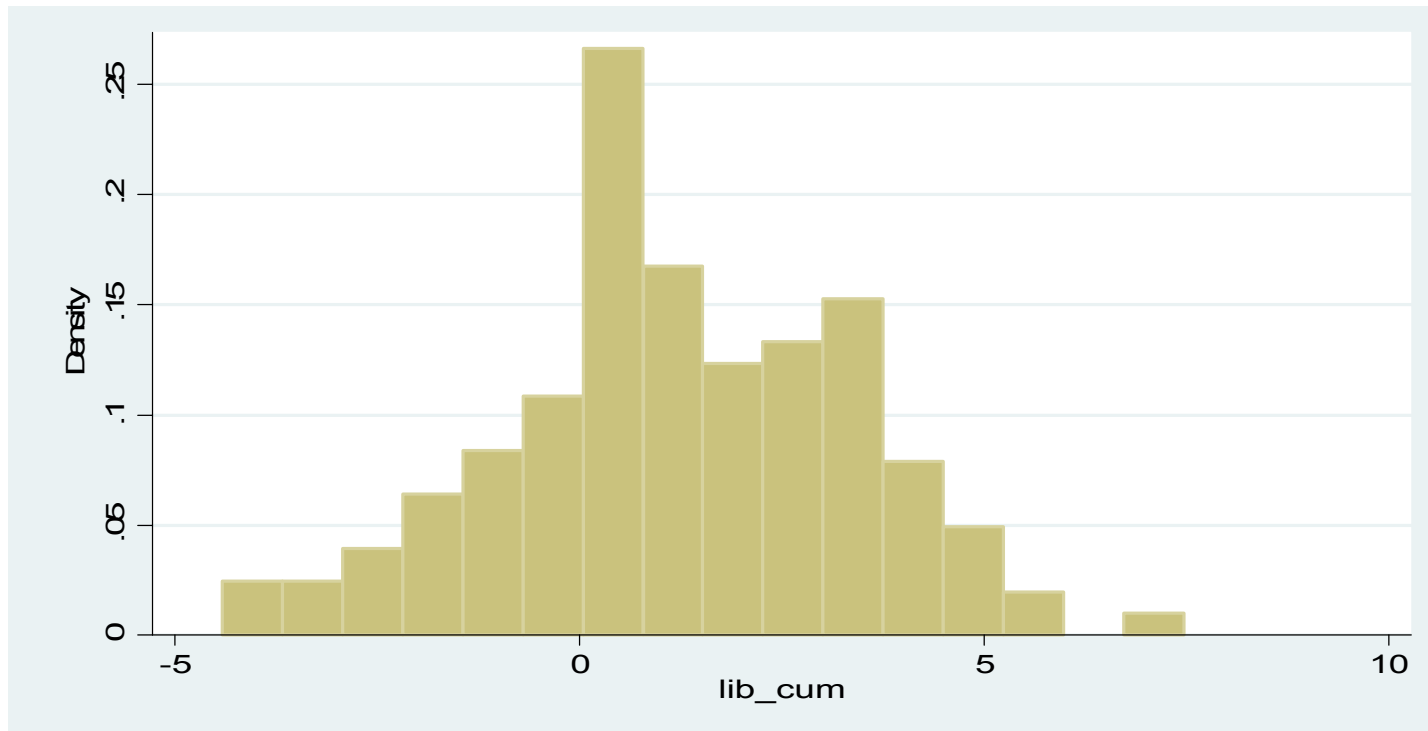
Histogram of the t-statistics of coefficients of structural reforms on economic growth: evidence from the 46 papers (Figure 1 in Babecký and Campos, 2011)

Link between structural reform and growth: short-run effect



Histogram of the t-statistics of coefficients of contemporaneous structural reforms on economic growth (Figure 2 in Babecký and Campos, 2011)

Link between structural reform and growth: long-run effect



Histogram of the t-statistics of coefficients of cumulative effect of structural reforms on economic growth (Figure 3 in Babecký and Campos, 2011)

Reform in Transition

- Consensus from econometric evidence is non-existent
- For this paper we extended the sample from 46 to 60 econometric studies (1996-2013), for TEs
- To measure reform effects, we constructed the partial correlation coefficient

Measuring **effect size**:

- 'reform elasticity' not available (specifications in logs and/or in levels, different units of measurement)
- t-statistics (captures significance)
- sign of the effect

We choose: Partial correlation coefficient
of the effect of reforms on growth (r)

$$r = \frac{t}{\sqrt{t^2 + df}}$$

$$SE(r) = r / t$$

t = *t-statistics*

SE = *standard error*

df = *degrees of freedom*

=> r characterizes both the magnitude and the significance of the effect;

Reform in Transition

What we did?

- We estimated the average effect constructing the partial correlation coefficient of the effect of reforms on growth (r), and corrected it for
- publication bias
- heterogeneity

Role of Time Horizon:

Figure 1: Reforms hurt in the short run, but spur long-run growth

Note: 537 coefficients from 60 papers

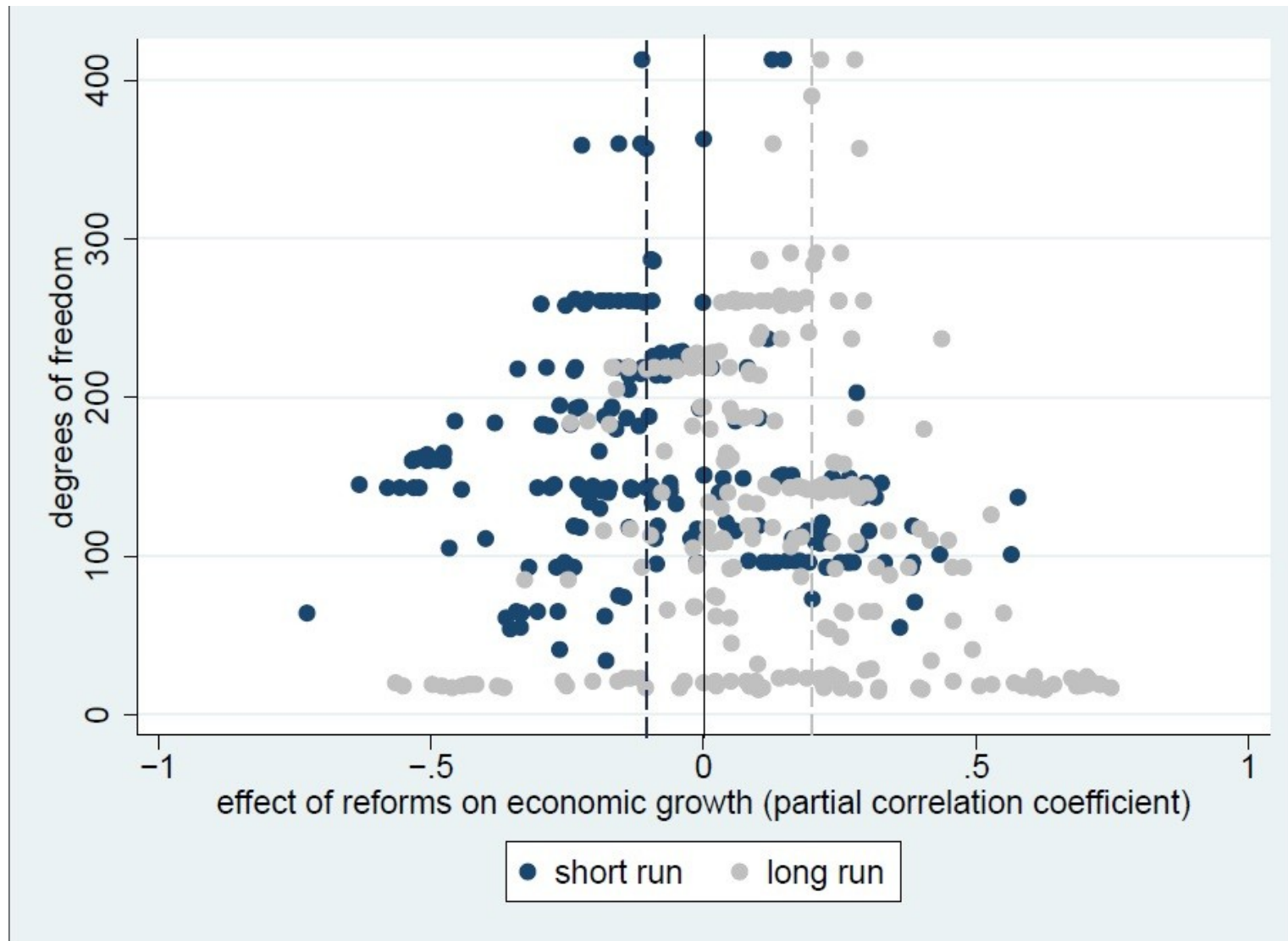


Table 2. Estimating the average reform effect

Method	Short run			Long run		
	Estimated effect	95% confidence interval		Estimated effect	95% confidence interval	
Simple average	-0.052	-0.084	-0.021	0.146	0.118	0.173
Fixed effects	-0.081	-0.091	-0.072	0.135	0.125	0.145
Random effects	-0.056	-0.087	-0.025	0.143	0.122	0.164

Notes: “Estimated effect” = partial correlation coefficient

“Simple average” = unweighted arithmetic average of all estimates

“Fixed effects” = the average weighted by the inverse of the standard error of the partial correlation coefficient (‘precision weighting’).

“Random effects” => additionally to FE, the underlying reform effects estimated in different models may vary

Are the estimated effects important? Guidelines (in economics):

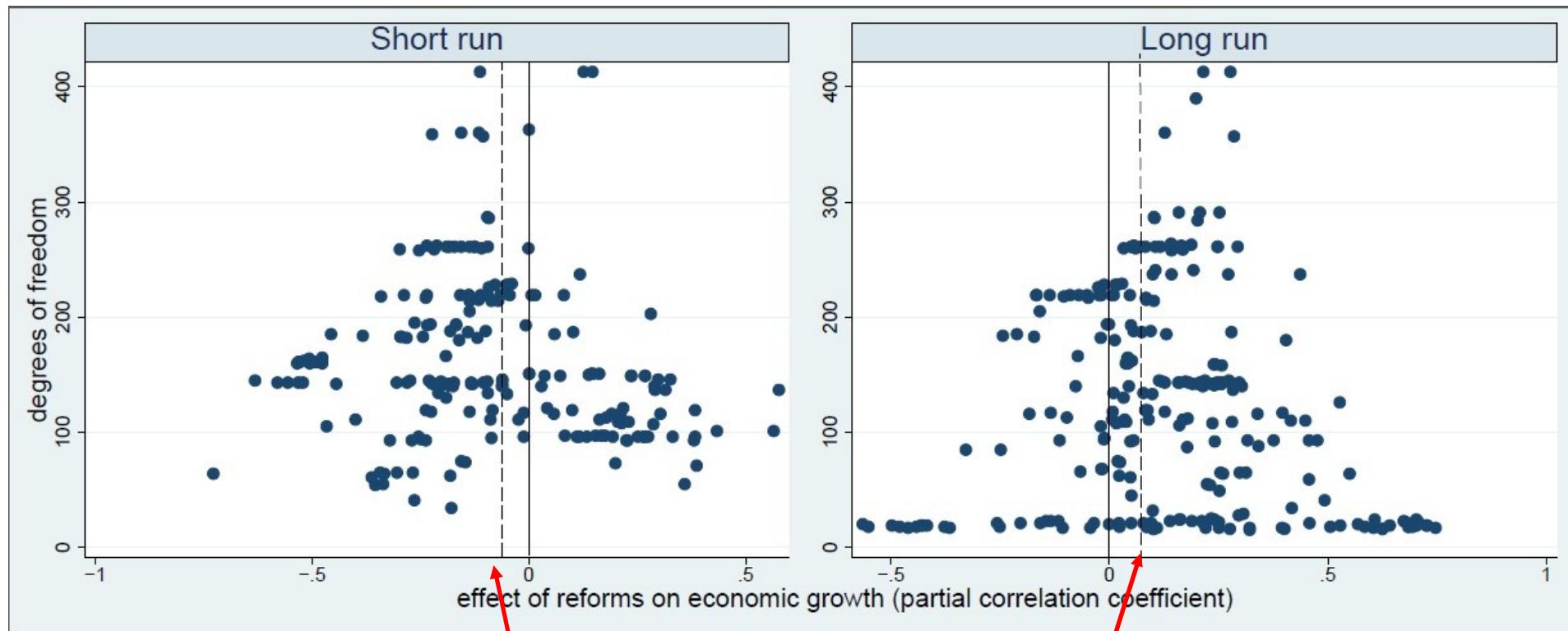
Doucouliaos (2011): $r < 0.07$ – no important effect

$0.07 < r < 0.17$ – small effect

$0.17 < r < 0.33$ – medium effect

$r > 0.33$ – strong effect

Figure 2. Funnel plots suggesting slight publication bias



Averages of all reported estimates

In the absence of PB, funnels should be symmetrical with respect to the averages

Test of publication bias

$$r_i = r_0 + \beta_0 \cdot Se(r_i) + u_i$$

	Short run			Long run		
	Fixed	Robust	Clustered	Fixed	Robust	Clustered
Publication bias (coef. β_0)	4.137 ^{***} (0.947)	4.179 ^{***} (0.961)	4.137 [*] (2.036)	0.313 (0.290)	0.265 (0.300)	0.313 (0.586)
Effect beyond bias (Constant)	-0.394 ^{***} (0.073)	-0.395 ^{***} (0.074)	-0.394 ^{**} (0.164)	0.110 ^{***} (0.025)	0.116 ^{***} (0.026)	0.110 [*] (0.056)
Observations	245	245	245	292	292	292

Publication bias:

=> Insignificant for the long run reform effects

=> Significant for the short run reform effects

Accounting for heterogeneity

$$r_i = r_0 + \beta_0 \cdot Se(r_i) + \gamma \cdot \text{Study design} + v_i$$

- Variable selection: BMA

Table 4: Explanatory variables

Variable	Description	Short run		Long run	
		Mean	Std. dev.	Mean	Std. dev.
prec	The precision of the estimated partial correlation coefficient (the inverse of the standard error)	12.666	2.675	10.535	4.456
<i>Data and methods</i>					
panel	= 1 if the model uses panel data.	0.996	0.064	0.750	0.434
endo	= 1 if model used is 2SLS, 3SLS, GMM, or cointegration.	0.298	0.458	0.274	0.447
fixed	= 1 if fixed effects estimation is used (or country dummies).	0.318	0.467	0.144	0.352
k	The number of explanatory variables.	13.449	10.478	10.048	9.768
start	The first year of the sample.	8.155	2.271	7.801	3.052
tspan	The number of years in the sample.	7.963	3.437	8.452	4.526
<i>Type of reform index</i>					
ebrd	= 1 if the reform index originates from the EBRD only.	0.453	0.499	0.620	0.486
comb	= 1 if a combination of EBRD and WB indices is used.	0.163	0.370	0.151	0.358
lii	= 1 if internal and/or price liberalization components are used as a reform measure.	0.069	0.255	0.048	0.214
lie	= 1 if external liberalization components are used.	0.069	0.255	0.045	0.207
lip	= 1 if privatization and banking reform components are used.	0.110	0.314	0.082	0.275
margeff	= 1 if <i>lii</i> , <i>lie</i> , and <i>lip</i> are used in the same specification.	0.118	0.324	0.068	0.253
av	= 1 if average (simple or weighted, or simple sum) of <i>lii</i> , <i>lie</i> , and <i>lip</i> is used.	0.645	0.480	0.798	0.402
cli	= 1 if the Cumulative Liberalization Index from the World Bank is used.	0.008	0.090	0.082	0.275
<i>Measure of dynamics</i>					
lagdep	= 1 if lagged dependent variable is used in the regression.	0.184	0.388	0.154	0.362
speed	= 1 if speed is the measure of reform.	0.241	0.428	0.205	0.405
lags	= 1 if both contemporaneous and lagged reform variables are used.	0.620	0.486	0.534	0.500
time	= 1 if time dynamics is controlled for.	0.167	0.374	0.195	0.397

Table 4: Explanatory variables (continued)

Variable		Description	Short run		Long run	
			Mean	Std. dev.	Mean	Std. dev.
<i>Specification characteristics</i>						
ic	= 1 if controlled for initial conditions.		0.718	0.451	0.791	0.407
ic12	= 1 if the first cluster and/or second cluster of initial conditions from de Melo <i>et al.</i> (1997) is used.		0.278	0.449	0.250	0.434
nic	The number of types controls for initial conditions.		1.624	1.916	1.740	1.637
stabil	= 1 if controlled for stabilization.		0.910	0.286	0.726	0.447
nstab	The number of types of controls for stabilization.		1.518	0.939	1.086	0.925
infl	= 1 if inflation is controlled for.		0.824	0.381	0.616	0.487
inst	= 1 if controlled for institutional development.		0.216	0.413	0.260	0.440
ninst	The number of types controls for institutional development.		0.229	0.449	0.411	0.871
fact	= 1 if controlled for factors of production.		0.294	0.456	0.229	0.421
nfact	The number of types controls for factors of production.		0.318	0.517	0.264	0.513
pubpr	= 1 if the study separates the effect of reform on public and private sector.		0.065	0.248	0.048	0.214
<i>Publication characteristics</i>						
journal	= 1 if the study is published in a refereed journal.		0.465	0.500	0.565	0.497
lgoog_pa	The logarithm of the number of citations per year from Google Scholar.		1.751	1.114	1.840	1.004
authaff	= 1 if all authors are from academia.		0.502	0.501	0.568	0.496

Table 5: Reform indicators

Reform type	<i>Reform index origin</i>	
	World Bank	EBRD
<i>lii</i>	Internal markets liberalization (liberalization of domestic prices and abolition of state trading monopolies)	Price liberalization
<i>lie</i>	External markets liberalization (liberalization of foreign trade and currency convertibility)	Trade and foreign exchange liberalization
<i>lip</i>	Private sector entry (small- and large-scale privatization and banking reform)	Small-scale privatization
		Banking reform and interest rate liberalization
		Large-scale privatization

Notes: Details on the World Bank indices are provided in [de Melo *et al.* \(1997\)](#). For the EBRD reform indicators, see the EBRD Transition Reports, which are available at yearly periodicity at www.ebrd.com.

Figure 3: BMA, model inclusion (short run)

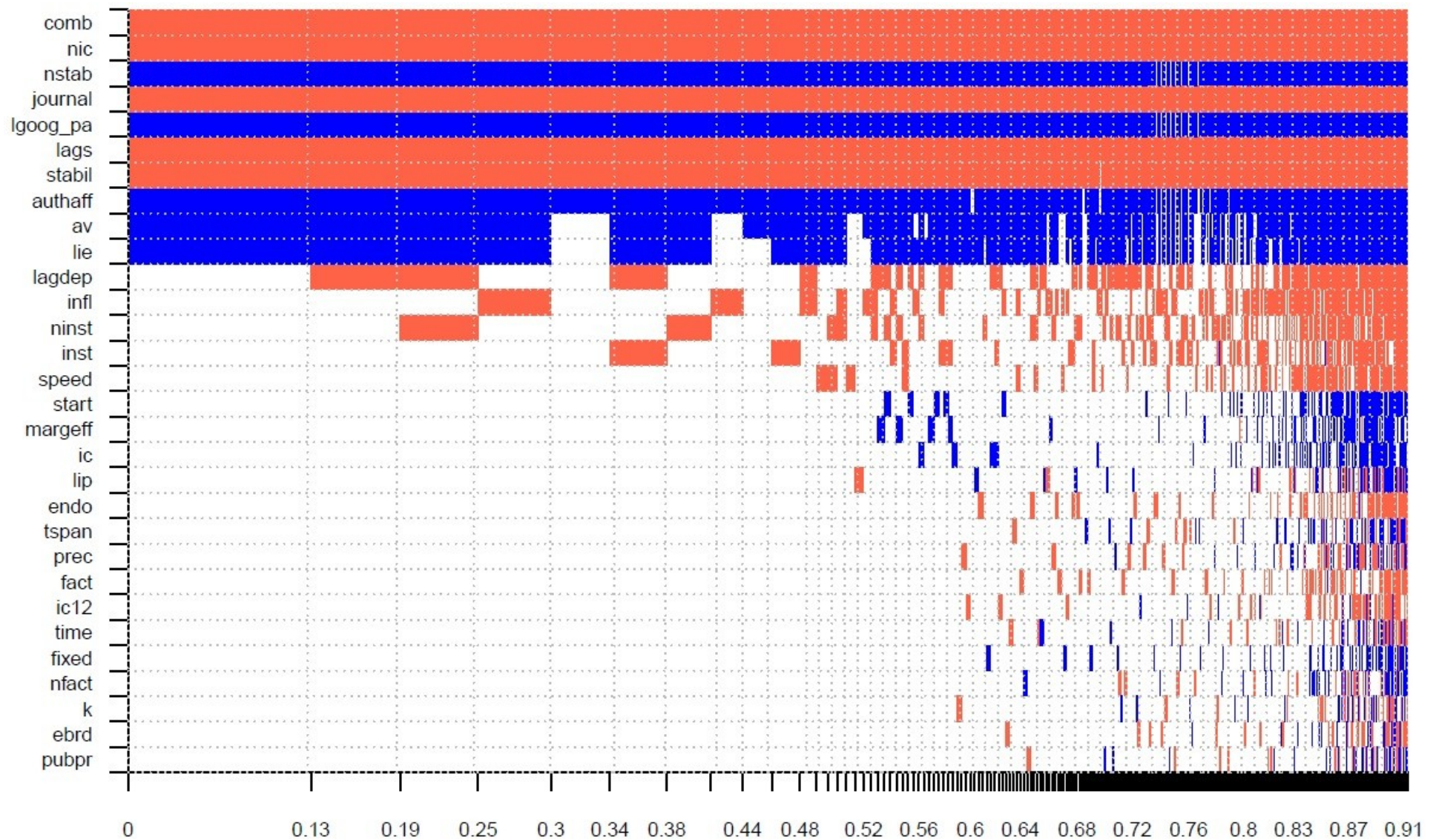
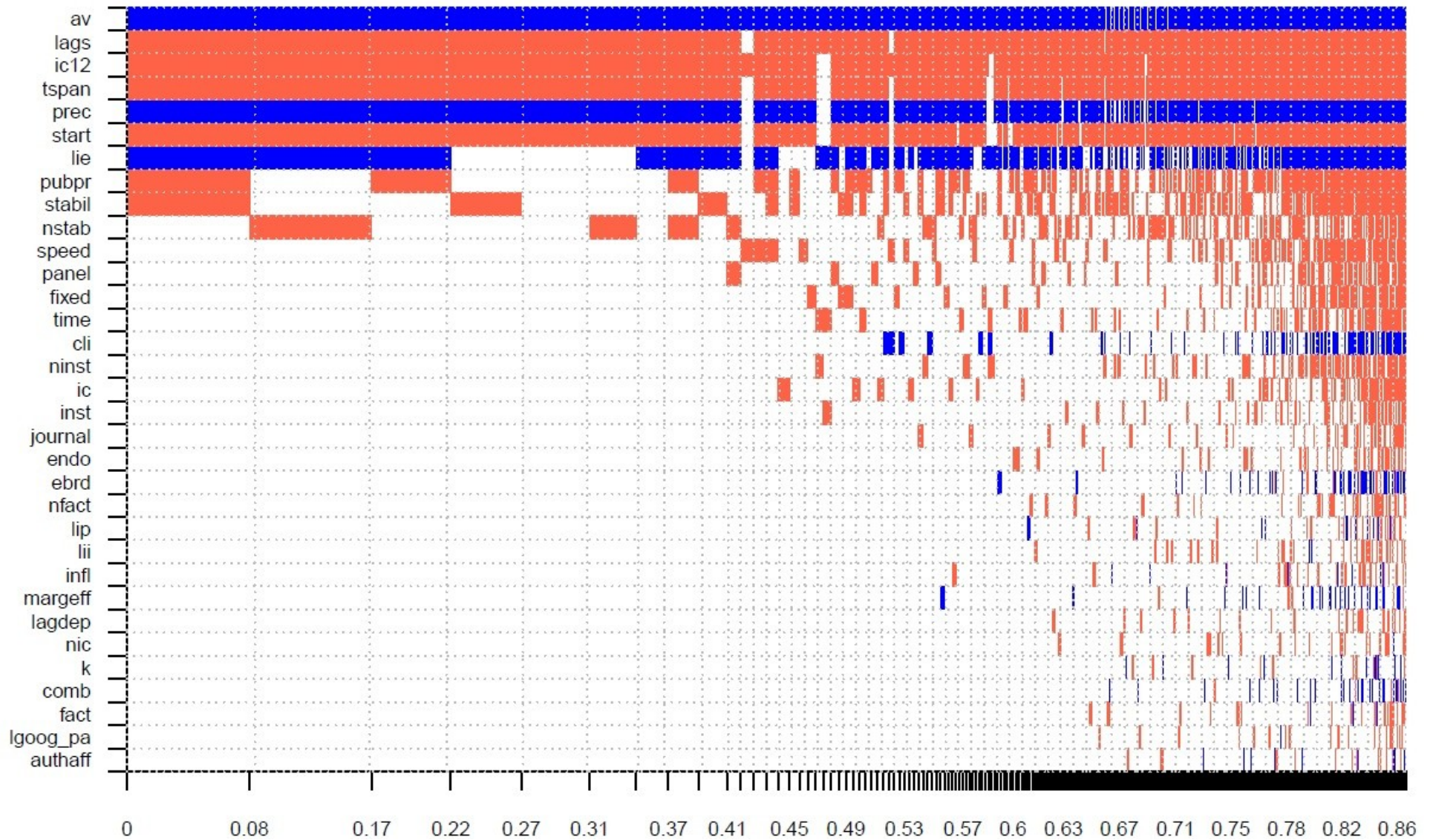


Figure 4: BMA, model inclusion (long run)



Accounting for heterogeneity: **Results**

$$r_i = r_0 + \beta_0 \cdot Se(r_i) + \gamma \cdot \text{Study design} + v_i$$

- Variable selection: BMA
- **Reform effect estimates:**
 - => **SR: -0.38** (-0.39..-0.40 if corrected for PB only)
 - => **LR: 0.27** (0.11..0.12 if corrected for PB only)

Conclusions

- Reviewed 60 studies published in 1996-2013. Constructed **partial correlation** coefficient to **capture both significance & magnitude** of the effect of reform on growth
- Short run costs, long run benefits: It takes time for the benefits of reforms to materialize
- The type of reforms matters: positive effects of external liberalization on growth, in both the short and long run

Future research

- External liberalization seems central: does it interact with other reforms? (reform complementarity)
- Exploring potential of these reforms
- Effects of other reforms – enterprise governance, competitiveness ... (as more empirical evidence becomes available)

Policy implications

- It is vital to improve understanding of how structural reforms influence economic growth (short-run costs versus long-run benefits)
- Our MRA results suggest that the initial costs of reform depend on the type of reform
- Need to account for structural reforms in macroeconomic modelling

References

- Babecký, J. and Campos, N. F. (2011): Does Reform Work? An Econometric Survey of the Reform-Growth Puzzle. *Journal of Comparative Economics*, 39(2), 140–158.
- Babecký, J. and Havránek, T. (2014): Structural Reforms and Growth in Transition: A Meta-Analysis. *Economics of Transition*, 22(1), 13–42.